

DSA 210 Final Report

How Online Fashion Trends Influence the Popularity of Fashion Products

Introduction

Fashion trends play an important role in shaping consumer behavior and product popularity in the clothing industry. Online fashion retailers continuously adapt their collections according to changing customer preferences, seasonal trends, and social media influence. As trends change rapidly, understanding the relationship between fashion popularity and product distribution has become increasingly important for both retailers and consumers. The purpose of this project is to analyze women's fashion product categories and investigate whether products associated with more popular fashion trends appear more frequently in the dataset. In order to explore this relationship, several data science techniques were applied, including data preprocessing, exploratory data analysis (EDA), machine learning classification, and Google Trends enrichment. The project focuses on selected women's fashion categories obtained from an online fashion dataset. By examining category frequencies, pricing patterns, and product naming structures, the project aims to better understand how fashion products are represented within the dataset. Additionally, a machine learning model was developed to predict product categories using only product names, allowing the project to explore whether textual information alone contains meaningful signals for category classification. Overall, this project combines statistical analysis, data visualization, machine learning techniques, and Google Trends enrichment to examine the relationship between fashion product characteristics and category popularity.

Dataset and Data Preprocessing

The dataset used in this project was obtained from Kaggle and contains women's fashion products collected from an online fashion retailer. Several different product categories were available in separate CSV files. For this project, four categories were selected because they were highly relevant to fashion trends and provided enough data for meaningful analysis. The selected categories were Dresses & Jumpsuits, Skirts, Tops & Bodysuits, and Knitwear. After selecting the categories, the separate datasets were merged into a single dataframe using the pandas library. Before combining the files, a category label column was added to each dataset in order to preserve category information after merging. The final combined dataset contains 1,038 products. Dresses & Jumpsuits represent the majority of the dataset with 643 products, while Skirts contain only 59 products. Knitwear and Tops & Bodysuits contain 228 and 108 products respectively. Several preprocessing operations were performed before the analysis stage. The automatically generated index column named "Unnamed: 0" was removed because it did not contain meaningful information. The price column also required cleaning because it originally included currency symbols, commas, and string formatting. The values were converted into numerical float format by removing unnecessary characters and whitespace. The preprocessing stage was important because clean and properly formatted data is essential for accurate analysis and model performance.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to better understand the structure of the dataset and identify important patterns between categories, prices, and product names. The first stage of the analysis focused on category distribution. The results clearly showed that Dresses & Jumpsuits dominate the dataset. Knitwear also represents a significant portion of the dataset, while Skirts contain the smallest number of products. This imbalance suggests that certain categories may be more popular or more heavily represented by the retailer. Dresses appear much more frequently than other categories, which may indicate stronger demand or a broader variety of products in that category. The second stage of the analysis focused on product prices. The average prices by category were calculated as follows: - Dresses & Jumpsuits: 4415.51 - Skirts: 3395.08 - Knitwear: 3318.07 - Tops & Bodysuits: 2654.81 The results showed that Dresses & Jumpsuits have the highest average price among the selected

categories, while Tops & Bodysuits have the lowest average price. Another part of the EDA involved keyword frequency analysis. Product names were analyzed using keyword matching in order to observe how frequently trend-related words appeared. Keyword counts: - “dress”: 572 - “knit”: 278 - “top”: 157 - “skirt”: 59 The keyword analysis strongly supported the earlier observations regarding category dominance. The word “dress” appeared significantly more often than the other selected keywords, reinforcing the idea that dresses are the most prominent category within the dataset. Google Trends data was also incorporated as an external enrichment source to compare real-world fashion search popularity with the dataset frequencies. The analysis showed that “dress” had the highest average search interest, which aligned with the dataset analysis where dresses were also the dominant category in terms of product count and keyword frequency.

Hypothesis and Machine Learning

The initial hypothesis of this project was that products associated with more popular fashion trends appear more frequently in the dataset. The exploratory analysis partially supported this hypothesis because dress-related products appeared much more frequently than the other selected categories. Dresses also showed the highest keyword frequency and the highest average price. In addition to the statistical analysis, a machine learning classification model was developed to predict product categories using product names. The product names were converted into numerical representations using CountVectorizer, which transforms text into word-frequency based features. After feature extraction, the dataset was divided into training and testing sets using an 80/20 split. A Multinomial Naive Bayes model was selected because it is commonly used for text classification problems and performs efficiently on word-frequency data. The model achieved approximately 90% accuracy on the test set. This means the model correctly predicted product categories in most cases. The model performed best on Dresses because dress-related product names contain very distinctive terms. The lowest performance was observed for Tops & Bodysuits because the word “top” is more general and overlaps with naming patterns from other categories. Despite some limitations, the results demonstrate that product names contain strong category-related information and can be effectively used for classification tasks.

Conclusion

This project investigated how online fashion trends may influence the popularity of fashion product categories using exploratory data analysis and machine learning techniques. The analysis showed that Dresses & Jumpsuits dominate the selected dataset both in product frequency and average price. Keyword analysis also revealed that dress-related terminology appears significantly more often than the other selected keywords. The machine learning stage demonstrated that product names alone contain enough information to successfully classify product categories. The classification model achieved approximately 90% accuracy, indicating strong predictive performance. Although the project has limitations such as dataset imbalance and limited category selection, the integration of Google Trends data provided external validation for the observed popularity patterns. Overall, the project successfully combined data preprocessing, exploratory data analysis, visualization, machine learning methods, and external trend enrichment to explore patterns in online fashion products.

Machine Learning Performance

Category	F1-Score
Dresses	0.94
Knitwear	0.84
Skirts	0.83
Tops	0.79

References

Kaggle Dataset:

Men and Women Clothing E-Commerce Dataset

Python Libraries:

- pandas
- matplotlib
- scikit-learn